



Part *of the* SYSTEM

No organism lives entirely alone. As well as all the living things around it—plants, animals, bacteria, and fungi—it interacts with the air, soil, water, and sunlight. We call such a group of organisms a **community** and, together with the environment it lives in, an **ecosystem**.



ECOSYSTEMS

Ecosystems can be as small as a crack in a rock or as big as Earth itself. Really large ecosystems are called biomes and consist of many smaller ecosystems.

Each ecosystem is made up of a number of habitats.

A habitat is the place where one or more species live. The habitat must be able to supply everything the organism needs or it will go somewhere more suitable.

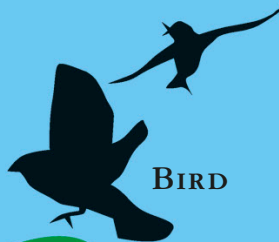
Fox



INSECT



BIRD



Ideal habitat

A tree provides a habitat for many species, including birds, insects, and mammals. Birds roost on branches, eat insects, and make nests to raise their young. Mammals might burrow among the tree's roots and eat nuts and seeds. Insects eat leaves and lay eggs on them. The tree relies on the bird to get rid of pests and the mammal to spread its seeds. Each organism has a different use for the tree, but they all interact with, and depend on, the other species that live there.

Finding your niche

Every organism has a way of life that is unique to its species. While many species may share a habitat, each has its own role in the community called a niche. For example, a forest may provide the habitat for a fox. Its niche

is that of a predator eating smaller animals that also live in the

forest. Out on a prairie, a coyote occupies the same predator niche as a fox. However, you would never find a fox and a coyote trying to occupy the same niche in the same habitat. There wouldn't be enough food for both species to survive.

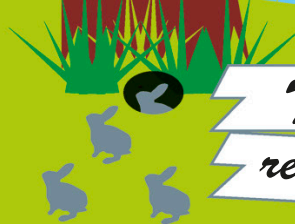
FOREST



Always changing

Ecosystems change over time. A piece of bare soil may eventually become a forest as new plants take root and attract animals that feed on them. They, in turn, attract predators. The ecosystem gradually becomes more complex as more species come in to fill new gaps. Eventually the ecosystem matures and reaches a balance point where all the things that live there have exactly what they need to survive.

RABBITS



COYOTE



GOOD-BYE!